

Ishaan Mittal

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EDUCATION

University of Waterloo

Candidate for Bachelor of Applied Science - BAsC, Computer Engineering

Waterloo, ON

Sept. 2025 – Present

EXPERIENCE

Quantum Hardware/Software Development Intern

SQE.io

March. 2026 – Present

East Greenwich, RI

- Implementing a **quantum safe financial transaction system** using the **QRS SDK**, **Python**, and **TypeScript**, enabling end-to-end **encrypted** payment workflows without security risks to emerging post quantum threats.
- Architecting a **quantum resistant communication platform** for encrypted text and image transmission using **Python**, **TypeScript**, and **TLS**, securing **sensitive data** against future quantum based attacks.
- Building a post quantum **industrial IoT system** using **ESP32**, **Raspberry Pi**, **RS485**, **Modbus**, and **Python** to enable next generation secure hardware communication through SQE-backend with **sub-500ms** latency.

Software Development Intern

Clouds Analytics

Feb. 2026 – May. 2026

Brampton, ON

- Developed secure **full-stack web applications** using **React**, **Python**, and **Flask** for cloud analytics platforms, improving usability and contributing to a **30%** increase in user adoption and sales.
- Applied **NIST-aligned** secure coding, **SOC 2 compliance**, **vulnerability assessments**, and **penetration testing** to reduce security risks by **40%** and strengthen application security.
- Collaborated in **agile** teams via **code reviews** and **debugging** to ensure on-time delivery.

Autonomy Software Developer

Waterloo Aerial Robotics Group

Sept. 2025 – Dec. 2025

Waterloo, ON

- Engineered a **Python-based ground station** using **MAVLink**, **multi-process (KPN) architecture**, and **computer vision**, improving drone communication reliability and orientation accuracy by **35%**.
- Collaborated with teams to integrate **communication** and **control modules** to improve **path accuracy**.
- Debugged** and **logged** code using **pytest** and **pylint** to achieve **100%** test coverage and maintainable code.

Cybersecurity Intern

NetraScale

Jul. 2025 – Sept. 2025

Toronto, ON

- Boosted **UI/UX** interactivity by **25%** by adding dashboards and visuals in **React** using **prompt engineering**.
- Optimized analytics workflows with **FastAPI** and **Chart.js**, developing **LLM-based** data visuals that improved team understanding of relationships between activity, traffic, and threats, reducing analysis time by **30%**.
- Simplified **20+ cybersecurity** policies into visual summaries, improving client clarity and compliance by **40%**.

PROJECTS

Greenify 🌱 | Python, JavaScript, OpenCV, Figma, ML

Oct. 2024 – Oct. 2024

- Developed a **Python ML** application to detect food waste, promoting **environmental sustainability**.
- Processed **900+** images with **YOLO** and **OpenCV** to build **ML** datasets that improved model accuracy.
- Designed **UI** in **Figma** and added interactivity with **CSS** and **JavaScript** to improve user experience.

SmartVault 🔒 | Java, OpenCV, JavaSwing, Alpha Vantage, Git

May 2024 – Jun. 2024

- Programmed a **Java-based** banking simulator with FaceID authentication, fund transfers, and stock tracking.
- Embedded **facial recognition** through **OpenCV** to enable secure, multi-factor user verification.
- Refined transfer/cashout systems and integrated **AlphaVantage API** for **10+** stocks to boost user engagement.

SKILLS

Languages: Java, Python, C#, C++, JavaScript, HTML/CSS, Typescript

Frameworks: React, Node.js, Express.js, FastAPI, Jupyter Notebook, OpenCV, JavaSwing

Developer Tools: Git, VS Code, IntelliJ, Claude, NetBeans, Figma, Replit, Docker, MongoDB, Linux

Hardware: Arduino, Verilog, Limelight, LaserCan, ESP32, FPGA, Oscilloscope, Raspberry Pi

CERTIFICATIONS

Certifications: Google Cybersecurity Foundations, Microsoft Security Essentials, Introduction to Cloud Computing, Microsoft Azure AI Essentials, Introduction to Generative AI, The Cybersecurity Threat Landscape